

Jesse Hoppo

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EDUCATION

University of Adelaide, SA

Feb 2023 – Nov 2026

- **Major:** Bachelor of Software Engineering (Honours)
- **Coursework:** Data Structures, Computer Networking, Artificial Intelligence, Cybersecurity, Distributed Systems, Operating Systems, Computer Systems

EXPERIENCE

Software Engineering Intern | Australian Energy Market Operator (AEMO)

Dec 2025 – Present

- Implemented automated data integrity checks using Git pre-commit hooks and R-based directory hashing, reducing manual verification time and preventing potential invalid data merges across SQL and Oracle pipelines.
- Optimized CI/CD workflows for internal codebases, reducing deployment failure rates and shortening release cycles using Docker and Azure DevOps.
- Refactored legacy SMTP scripts into a centralized shared library for wider operational use, reducing manual interaction and saving ~5 engineering hours per week via automated reporting processes.
- Developed and enhanced R Shiny dashboards and energy demand models to meet evolving stakeholder requirements.
- Authored 3+ technical documents and onboarding guides (Git, Networking, Azure AD), accelerating onboarding processes for new interns/graduates and resolving onboarding issues faster.
- Led deployment of ELK stack (Filebeat, Journald, Logstash) across RHEL environments using Podman and Pyinfra, enabling real-time observability for production and development services.

Heart Failure Portal (POC Prototype) | Capstone Project / CEIH Heart Failure Sub-committee

Feb 2026 – Present

- Led agile delivery as team Scrum Lead using Jira, coordinating sprint planning, stand-ups, task prioritization, and retrospective reviews to ensure timely POC delivery.
- Evaluated and documented justification of Wagtail CMS (Python/Django) based on system requirements, considering scalability, security, accessibility, and ease of use.
- Designed system workflows and data structures to support content management and user interactions within the portal.
- Developed and configured CMS components to meet functional requirements, including content models and basic database interactions.
- Designed a risk management framework to proactively identify, monitor, and mitigate delivery risks.

Laboratory Technician | Australian Geotechnical Testing (AGT)

Apr 2025 – Dec 2025

- Collaborated with civil engineers and field technicians to deliver reliable geotechnical testing services.
- Performed material testing (MMD, CBR, PI, PSD) on sand, clay, and quarry samples per Australian Standard AS1289.
- Trained new employees in MDD testing, ensuring adherence to regulatory guidelines and company standards.

PROJECTS

Simplified Redis | C/C++, Networking, Database

- Built a Redis-like in-memory database via C++ with a TCP socket-based communication layer to support client-server interaction.
- Designed custom networking using IPv4 with buffered I/O for reliable bidirectional communication.
- Implemented hash-table based storage with performance optimizations including cache alignment, stack allocation, and loop unrolling increasing operational speed over 50%.
- Focused on low-latency operations and memory efficiency to simulate real-world use and database internals.

Medical Dashboard | C++, Qt, Git, Jira

- Developed a cross-platform application using C++ and Qt to simulate online-based medical consultation and use workflows including prescription ordering, appointment booking, user-health overviews, and more.
- Led documentation workflows, providing structured README guides and robust Confluence pages to support the full development lifecycle.
- Researched and applied WCAG accessibility standards to ensure development meets sector regulations and improves the user experience.
- Collaborated with team members in an Agile environment using Git and Jira Kanban to manage the development lifecycle.

Distributed API | Java, Bash, HTTP, Gson/JSON, Sockets, UML

- Designed and implemented a distributed aggregation server in Java supporting concurrent HTTP GET/PUT/DELETE requests.
- Implemented Lamport clocks to ensure consistency and ordering across nodes under concurrent updates.
- Built custom HTTP parsing and response handling to support key status code (200,201,204,400,500).
- Developed fault-tolerant architecture with crash-safe request handling with workflows and UML diagrams to support technical decisions.

SKILLS AND KNOWLEDGE

- **Languages:** C/C++, Python, R, Java
- **Systems & Backend:** Distributed systems, Networking, Concurrency, Low-level Memory Optimization
- **DevOps & Infrastructure:** Docker, Podman, CI/CD (Azure DevOps), ELK, Linux (RHEL, Ubuntu)
- **Data & Tools:** Dagster, Pyinfra, SQL, R-Shiny
- **Security:** GDB, Ghidra, BurpSuite, Nmap, Metasploit, Web Exploitation (XSS, CSRF), Memory Exploitation (Stack Smashing, Overflow)
- **Software:** Git/GitHub, Jira, Confluence, Wireshark